THE CHARACTER STRUCTURE OF THE COLLISION FLUCTUATION

ALEXANDER S. PETTY

Abstract. The truncated collision fluctuation sum $\Phi_x(s,\ell) = \sum_{b<p\le x} \Delta_p(\ell)\, p^{-s}$ appears to grow like $-\mu_\ell \cdot \log\log x$ at $s=1$. We present four observations. (1) Computation in bases 3, 6, 7, 10, and 12 is consistent with universality of this growth law. (2) The truncated sum decomposes over Dirichlet characters modulo b (proved). (3) The character components do not appear to be proportional to classical L-values (numerical evidence). (4) The per-frequency eigenvalue fluctuation sums appear to converge to nearly the same value (numerical evidence).

1. Introduction

This paper decomposes the collision fluctuation over Dirichlet characters.

The preceding paper [1] proved that the collision fluctuation sum $\Phi_x(s,\ell)$ converges for $\mathrm{Re}(s) > 2$ and conjectured logarithmic divergence at $s=1$:

$$\Phi_x(1,\ell) = \sum_{p>b} \frac{\Delta_p(\ell)}{p} \sim -\mu_\ell \cdot \log\log x.$$

The present paper examines the structure of this divergence.

2. Universality Across Bases

Conjecture 1 (Universality)**.** *The Mertens growth law $\Phi_x(1,\ell) \sim -\mu_\ell \cdot \log\log x$ and the convergence of $\Phi_x(s,\ell)$ for $s>1$ are independent of the base b.*

This conjecture is supported by computation in bases 3, 6, 7, 10, and 12, using all primes up to 8000.

The gate width ($b-1$ derangements) and the variance scaling (standard deviation $\sim \sqrt{Q}$) appear to be base-independent in the tested data. The same computations place the apparent critical boundary at $s=1$ in every base tested. The base determines the quantitative

Date: October 2022 (revised April 2026).

2020 *Mathematics Subject Classification.* 11A63, 11N05, 11M41.

base b	$b-1$	$\Phi_x(1,1)$	$\Phi_x(0.9,1)$	behavior at $s=0.9$
3	2	-1.38	diverging	$\uparrow$
6	5	-1.29	diverging	$\uparrow$
7	6	-1.34	diverging	$\uparrow$
10	9	-1.37	diverging	$\uparrow$
12	11	-1.32	diverging	$\uparrow$

TABLE 1. $\Phi_x(1,1)$ at $x = 8000$ grows logarithmically in every base tested. $\Phi_x(0.9,1)$ diverges as a power law. In the tested data, the apparent critical boundary is $s = 1$ in every base.

details (the Mertens constant μ_ℓ and the variance constant λ_b) but not the observed qualitative structure.

3. Decomposition over Dirichlet Characters

The primes coprime to b partition into $\phi(b)$ spectral classes by their residue modulo b. The Dirichlet characters χ modulo b are the natural basis for functions on these classes [2, 3].

Definition 2. The *truncated character-weighted fluctuation sum* for character χ modulo b is

$$L_{\chi,x}(s,\ell) = \sum_{b<p\le x} \frac{\chi(p)\,\Delta_p(\ell)}{p^s}.$$

Proposition 3 (Character decomposition)**.** *The truncated fluctuation at primes in spectral class r is*

$$F_{r,x}(s,\ell) = \sum_{\substack{b<p\le x\\ p\equiv r \bmod b}} \frac{\Delta_p(\ell)}{p^s},$$

and it satisfies the exact finite identity

$$F_{r,x}(s,\ell) = \frac{1}{\varphi(b)} \sum_{\chi} \overline{\chi}(r)\, L_{\chi,x}(s,\ell).$$

The total truncated fluctuation sum is

$$\Phi_x(s,\ell) = \sum_r F_{r,x}(s,\ell) = L_{\chi_0,x}(s,\ell),$$

the trivial-character component.

Proof. For each prime $p \nmid b$, the standard orthogonality relation

$$\frac{1}{\varphi(b)} \sum_{\chi} \overline{\chi}(r)\chi(p) = \begin{cases} 1, & p \equiv r \pmod{b}, \\ 0, & p \not\equiv r \pmod{b} \end{cases}$$

isolates the residue class of p. Multiplying by $\Delta_p(\ell)\, p^{-s}$ and summing over $b < p \leq x$ gives

$$\frac{1}{\varphi(b)} \sum_{\chi} \overline{\chi}(r) L_{\chi,x}(s,\ell) = \sum_{b<p\leq x} \left(\frac{1}{\varphi(b)} \sum_{\chi} \overline{\chi}(r)\chi(p) \right) \frac{\Delta_p(\ell)}{p^s} = F_{r,x}(s,\ell).$$

Summing over all residue classes coprime to b leaves only the trivial character, since $\sum_r \overline{\chi}(r) = 0$ for nontrivial χ and equals $\varphi(b)$ for χ_0. □

Remark 4. For $\mathrm{Re}(s) > 2$, the preceding paper [1] proves convergence of $\Phi_x(s,\ell)$ as $x \to \infty$. In that domain the same orthogonality identity passes to the limit, so the infinite sums decompose in the same way.

4. The Character Components

Conjecture 5 (Component structure)**.** *Each truncated character component $L_{\chi,x}(1,\ell)$ appears to diverge individually with a character-specific Mertens constant.*

This is consistent with computation in base 10 at lag $\ell = 1$ using all primes up to 10000.

primes	$\lvert L_{\chi_0,x}\rvert$	$\lvert L_{\chi_1,x}\rvert$	$\lvert L_{\chi_2,x}\rvert$	$\lvert L_{\chi_3,x}\rvert$
100	1.073	0.254	0.321	0.254
500	1.278	0.356	0.350	0.356
800	1.341	0.380	0.355	0.380
1225	1.393	0.398	0.350	0.398

Table 2. Truncated character component magnitudes at $s = 1$. The trivial character χ_0 carries the dominant weight. $|L_{\chi_1,x}| = |L_{\chi_3,x}|$ by conjugate symmetry.

The trivial character χ_0 carries the largest truncated weight ($|L_{\chi_0,x}| \approx 1.4$ at the tested cutoff). The non-trivial characters contribute smaller but nonzero corrections ($|L_{\chi_j,x}| \approx 0.35$–$0.40$).

5. Relation to Classical L-Values

The truncated character components $L_{\chi,x}$ do not appear to be proportional to classical Dirichlet L-values.

Remark 6. In base 10 at lag $\ell = 1$, the ratios $L_{\chi_2,x}(1,1)/L(1,\chi_2) \approx 0.54$ and $|L_{\chi_1,x}(1,1)|/|L(1,\chi_1)| \approx 0.40$ are not simple rational numbers and depend on the lag ℓ. This suggests that the character components $L_{\chi,x}(s,\ell)$ are not reducible to classical Dirichlet L-functions [2, 4].

6. Spectral Equidistribution

For primes p where b is a primitive root, the cross-alignment matrix is permutation-similar to a circulant. Its eigenvalues are therefore determined by the autocorrelation profile through the discrete Fourier formulas developed in the spectral-structure and autocorrelation papers. Computation suggests that, as $p \to \infty$, the normalized eigenvalue λ_k/Q converges to

$$\lambda_k/Q \;\longrightarrow\; \frac{b^2 \sin^2(\pi k/b)}{(\pi k)^2}.$$

Define the eigenvalue fluctuation $\psi_p(k) = \lambda_k/Q - b^2 \sin^2(\pi k/b)/(\pi k)^2$, and the frequency fluctuation sum

$$G(s,k) = \sum_{p>b} \frac{\psi_p(k)}{p^s}.$$

Conjecture 7 (Spectral equidistribution)**.** *Each $G(1,k)$ converges at $s = 1$. Moreover, the limiting values are approximately equal across all frequencies: $G(1,k) \approx 0.165$ for $k = 1,\ldots,5$ in base 10.*

primes	$G(1,1)$	$G(1,2)$	$G(1,3)$	$G(1,4)$	$G(1,5)$
100	0.1642	0.1642	0.1642	0.1640	0.1639
500	0.1650	0.1650	0.1649	0.1646	0.1644
1000	0.1651	0.1651	0.1650	0.1646	0.1644

Table 3. The per-frequency fluctuation sums converge to nearly the same value for all k in the tested range. The eigenvalue fluctuations appear to distribute almost uniformly across frequencies.

The near-equality of $G(1,k)$ across frequencies suggests that the collision fluctuations are not concentrated at any particular harmonic mode.

7. Polarity Equidistribution (Observational)

The cumulative polarity of the repetend traces a path on the fixed circle $\mathbb{Z}/(b-1)\mathbb{Z}$. At each position of the repetend, the running digit sum modulo $b-1$ takes a value in $\{1, \ldots, b-1\}$. The distribution of these values is the *polarity histogram*: the vector $(h_1, \ldots, h_{b-1})$ where h_j counts how many repetend positions have running digit sum congruent to j modulo $b-1$.

For primes with b as primitive root, the polarity histogram appears to approach the uniform distribution $1/(b-1)$ as $p \to \infty$. The path visits each value approximately $L/(b-1)$ times. The polarity circle is the limiting distribution: computation suggests the field approaches equidistribution on it.

The collision fluctuations measure departure from this limiting distribution. The Mertens growth law asserts that these departures, summed over primes with weight $1/p$, accumulate at rate $\log\log x$. The spectral equidistribution asserts that the departures are flat across frequencies. Together, these computations suggest that the field approaches its equilibrium on the fixed polarity circle, and that the rate of approach is controlled by the critical boundary $s = 1$.

8. Remarks

The landscape. The collision fluctuation now has four layers of structural analysis: universality across bases, character decomposition with independent components, new arithmetic objects distinct from classical L-values, and spectral equidistribution of eigenvalue fluctuations. The fluctuation decomposes along three orthogonal axes (spectral class, frequency, polarity), and the computations are consistent with the same boundary behavior along each axis.

Open directions. The character decomposition reduces the study of the Mertens growth law to $\phi(b)$ independent problems: one for each L_χ at $s = 1$. Whether any of these components admit a closed-form Mertens constant, and whether the spectral equidistribution has a theoretical explanation, are open questions. The collision parameter $c = b(1 - b^\ell)^{-1} \bmod p$ (the residue class that determines whether the multiplier b^ℓ deranges the bin partition) and its equidistribution across spectral classes provide the analytic handle.

A modified fluctuation sum, corrected for the per-class bias of S, may converge at $s = 1$. If so, that centered sum would be the collision analog of the prime number theorem.

References

[1] A. S. Petty, The collision fluctuation sum, research note, 2024.
[2] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Springer, 2000.
[3] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, AMS Colloquium Publications, vol. 53, 2004.
[4] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge University Press, 2007.

Email address: `alexander.petty@gmail.com`